

# Algebraic Coding Theory

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## Abstract

Notes taken while studying Algebraic Coding Theory, mostly from Hill book [1] and Lint book [2].

Usually while reading books and papers I take handwritten notes in a notebook, this document contains some of them re-written to *LaTeX*.

The proofs may slightly differ from the ones from the books, since I try to extend them for a deeper understanding.

## Contents

|          |                                                                    |           |
|----------|--------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Vector spaces over finite fields, quick recap</b>               | <b>2</b>  |
| <b>2</b> | <b>Linear codes</b>                                                | <b>3</b>  |
| 2.1      | Some definitions . . . . .                                         | 3         |
| 2.2      | Equivalence . . . . .                                              | 4         |
| 2.3      | Encoding with a linear code . . . . .                              | 4         |
| 2.4      | Decoding with a linear code . . . . .                              | 5         |
| 2.4.1    | Cosets . . . . .                                                   | 5         |
| 2.5      | Probability of error correction (in binary linear codes) . . . . . | 6         |
| 2.6      | Exercises . . . . .                                                | 7         |
| <b>3</b> | <b>Dual code, PCM and Syndrome</b>                                 | <b>11</b> |
| 3.1      | Dual code . . . . .                                                | 11        |
| 3.2      | PCM (parity-check matrix) . . . . .                                | 13        |
| 3.3      | Syndrome . . . . .                                                 | 14        |
| 3.4      | Incomplete decoding . . . . .                                      | 16        |
| 3.5      | Exercises . . . . .                                                | 16        |
| <b>4</b> | <b>Hamming Codes</b>                                               | <b>19</b> |
| 4.1      | Extended binary Hamming codes . . . . .                            | 20        |
| 4.2      | q-ary Hamming codes . . . . .                                      | 21        |

# 1 Vector spaces over finite fields, quick recap

Let  $q$  be a prime power.

$V(n, q)$ : set  $GF(q)^n$  of all ordered  $n$ -tuples over  $GF(q)$ .  $V(n, q)$  is a vector space.

A set of vectors  $\{v_1, v_2, \dots, v_r\}$  is said to be *linear dependent* if  $\exists$  scalars  $a_1, \dots, a_r$ , not all zero, such that

$$a_1v_1 + a_2v_2 + \dots + a_rv_r = 0$$

They are *linear independent* if

$$a_1v_1 + a_2v_2 + \dots + a_rv_r = 0 \xRightarrow{\text{implies}} a_1 = a_2 = \dots = a_r = 0$$

Let  $C$  a subspace of  $V(n, q)$ . Then a subset  $\{v_1, v_2, \dots, v_r\} \subseteq C$  is called a *generating set* (or *spanning set*) of  $C$  if every vector in  $C$  can be expressed as a linear combination of  $v_1, \dots, v_r$ .

A generating set of  $C$  which is also linearly independent is called a *basis* of  $C$ .

**Theorem 4.3.** Suppose  $\{v_1, v_2, \dots, v_k\}$  is a basis of a subspace  $C$  of  $V(n, q)$ . Then

- i. every vector of  $C$  can be expressed uniquely as a linear combination of the basis vectors.
- ii.  $C$  contains exactly  $q^k$  vectors.

*Proof.* i. suppose vector  $x \in C$  such that

$$x = a_1v_1 + a_2v_2 + \dots + a_kv_k$$

$$x = b_1v_1 + b_2v_2 + \dots + b_kv_k.$$

then

$$(a_1 - b_1)v_1 + (a_2 - b_2)v_2 + \dots + (a_k - b_k)v_k = 0.$$

But the set  $\{v_1, v_2, \dots, v_k\}$  is linearly independent, and so

$$a_i - b_i = 0 \quad \text{for } i = 1, 2, \dots, k,$$

i.e.,

$$a_i = b_i \quad \text{for } i = 1, 2, \dots, k.$$

- ii. By (i), the  $q^k$  vectors

$$\sum_{i=1}^k a_i v_i \quad (a_i \in GF(q))$$

are precisely the distinct vectors of  $C$ .

□

## 2 Linear codes

The theorems of this section are from [1] and follow the original enumeration of the book.

Note about notation:

A  $q$ -ary linear code  $[n, k, d]$ -code is also a  $q$ -ary  $(n, q^k, d)$ -code, but the opposite is not always true.

### 2.1 Some definitions

Let  $q$  be a prime power. View  $(F_q)^n$  as the vector space  $V(n, q)$ .

**Definition** (Linear code over  $GF(q)$ ). A *linear code over  $GF(q)$*  is a subspace of  $V(n, q)$ , with  $n \geq 0$ .

Thus  $C \subseteq V(n, q)$  is a linear code iff

- i.  $u + v \in C \quad \forall u, v \in C$
- ii.  $au \in C \quad \forall u \in C, a \in GF(q)$

If  $C$  is a  $k$ -dimensional subspace of  $V(n, q)$ , then the linear code  $C$  is called an  $[n, k]$ -code. (sometimes with minimum distance  $d$ :  $[n, k, d]$ -code).

**Definition** (Weight). Weight,  $w(x)$ , of a vector  $x \in V(n, q)$  is the number of non-zero entries of  $x$ .

**Lemma 5.1.** if  $x, y \in V(n, q)$ , then  $d(x, y) = w(x - y)$ .

*Proof.* the vector  $x - y$  has non-zero entries in precisely those places where  $x$  and  $y$  differ.  $\square$

**Theorem 5.2.** let  $C$  a linear code, let  $w(C)$  be the smallest of weights of the non-zero codewords of  $C$ . Then  $d(C) = w(C)$ .

*Proof.*  $\exists x, y \in C$  s.th.  $d(C) = d(x, y)$ .

By Lemma 5.1,  $d(C) = w(x - y) \geq w(C)$ , since  $x - y$  is a codeword of the linear code  $C$  (ie.  $x - y \in C$ ).

On the other hand, for  $x \in C$ ,  
 $w(C) = w(x) = d(x, 0) \geq d(C)$ , since  $0 \in C$ .

Hence,  $d(C) \geq w(C)$  and  $w(C) \geq d(C) \implies d(C) = w(C)$ .  $\square$

**Definition** (Generator matrix (GM)). a  $k \times n$  matrix whose rows form a basis of a linear  $[n, k]$ -code is called a *generator matrix* (GM) of the code.

## 2.2 Equivalence

**Theorem 5.4.** two known  $k \times n$  matrices generate equivalent linear  $[n, k]$ -codes over  $GF(q)$  if one matrix can be obtained from the other by a sequence of operations of the following types:

- R1. permutation of the rows
- R2. multiplication of a row by a non-zero scalar
- R3. addition of a scalar multiple of one row to another
- C1. permutation of the columns
- C2. multiplication of any column by a non-zero scalar

**Theorem 5.5.** let  $G$  a GM of an  $[n, k]$ -code. Then by performing the previous operations,  $G$  can be transformed to the standard form  $[I_k | A]$  where  $I_k$  is the  $k \times k$  identity matrix, and  $A$  is a  $k \times (n - k)$  matrix.

## 2.3 Encoding with a linear code

Let  $C$  a  $[n, k]$ -code over  $GF(q)$  with GM  $G$ .  $C$  contains  $q^k$  codewords  $\implies$  can be used to communicate any one of  $q^k$  distinct messages.

Identify these messages with the  $q^k$  tuples of  $V(k, q)$ .

Encoding:

msg vector:  $u = u_1 u_2 \dots u_k$ ; rows of  $G$ :  $r_1 r_2 \dots r_k$ . Encode

$$uG = \sum_{i=1}^k u_i r_i \in C$$

ie.  $uG$  is a codeword of  $C$ ; a linear combination of the rows of the GM.

Encoding function:

$$\begin{aligned} V(k, q) &\longrightarrow C \subseteq V(n, q) \\ u &\longmapsto uG \end{aligned}$$

If  $G$  is in standard form (5.5),  $G = [I_k | A]$  with  $A = [a_{ij}]$ , then encoding  $u$  is:

$$x = uG = x_1 x_2 \dots x_k x_{k+1} \dots x_n$$

where

$$\begin{aligned} x_i &= u_i \quad \text{for } 1 \leq i \leq k, \text{ message digits} \\ x_{k+i} &= \sum_{j=1}^k a_{ij} u_j \quad \text{for } 1 \leq i \leq n - k, \text{ check digits} \end{aligned}$$

the check digits add redundancy to give protection against noise.

## 2.4 Decoding with a linear code

Codeword  $x = x_1x_2 \dots x_n$ , received vector  $y = y_1y_2 \dots y_n$ , error vector  $e = y - x = e_1e_2 \dots e_n$ .

### 2.4.1 Cosets

Cosets here are similar to the ones from typical group theory, just from another perspective.

**Definition** (Coset). Let  $C$  a  $[n, k]$ -code over  $GF(q)$  and  $a \in V(n, q)$ . Then the set  $a + C = \{a + x | x \in C\}$  is called a *coset* of  $C$ .

**Definition** (Coset leader). the vector having minimum weight in a coset is called the *coset leader*.

**Lemma 6.3.** Let  $a + C$  a coset of  $C$  and  $b \in a + C$ . Then  $b + C = a + C$ .

*Proof.* since  $b \in a + C \implies b = a + x$  for some  $x \in C$ . If  $b + y \in b + C$  then

$$b + y = (a + x) + y = a + (x + y) \in a + C$$

For the other direction, if  $a + x \in a + C$  then

$$a + x = (b - x) + z = b + (z - x) \in b + C$$

hence  $a + C \subseteq b + C$ .

Therefore  $b + C = a + C$ . □

**Theorem 6.4** (Lagrange). suppose  $C \subseteq V(n, q)$  is a  $[n, k]$ -code over  $GF(q)$ . Then

- i. every vector of  $V(n, q)$  is in some coset of  $C$
  - ii. every coset contains exactly  $q^k$  vectors
  - iii. two cosets either are disjoint or coincide (partial overlap is impossible)
- Proof.* i. if  $a \in V(n, q)$  then  $a = a + 0 \in a + C$
- ii. the map

$$\begin{aligned} C &\longrightarrow a + C \quad \forall x \in C \\ x &\longmapsto a + x \end{aligned}$$

is a one-to-one map, hence  $|a + C| = |C| = q^k$ .

- iii. by contradiction, suppose cosets  $a + C$ ,  $b + C$  overlap. Then for some vector  $v$ ,

$$v \in (a + C) \cap (b + C)$$

thus for some  $x, y \in C$ ,  $v = a + x = b + y$ .

Hence  $b = a + (x - y) \in a + C$ , thus by Lemma 6.3 we have  $b + C = a + C$ . □

**Corollary . 6.4** shows that  $V(n, q)$  is partitioned into disjoint cosets of  $C$ :

$$V(n, q) = (0 + C) \cup (a_1 + C) \cup \dots \cup (a_s + C)$$

where  $s = q^{n-k} - 1$ , and by Lemma 6.3 we may take  $0, a_1, \dots, a_s$  to be coset leaders.

**Example 6.5.** Let  $C$  be the binary  $[4, 2]$ -code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

ie.  $C = \{0000, 1011, 0101, 1110\}$ .

Then the cosets of  $C$  are

$$0000 + C = C \text{ itself}$$

$$1000 + C = \{1000, 0011, 1101, 0110\}$$

$$0100 + C = \{0100, 1111, 0001, 1010\} = 0001 + C$$

$$0010 + C = \{0010, 1001, 0111, 1100\}$$

by Lemma 6.3:  $0001 \in 0100 + C$ , thus  $0001 + C = 0100 + C$ .

## 2.5 Probability of error correction (in binary linear codes)

To evaluate how good a code is, must calculate the *probability* that a received vector will be decoded as the codeword which was sent.

Let  $p$  be the symbol error probability.

Probability that the error vector is agiven vector of weight  $i$  is  $p^i(1-p)^{n-i}$ .

**Theorem 6.7.** Let  $C$  a binary  $[n, k]$ -code, and for  $i = 0, 1, \dots, n$  let  $\alpha_i$  denote the number of coset leaders of weight  $i$ .

Then the probability  $P_{corr}(C)$  that a received vector decoded by means of a standard array is the codeword which was sent is

$$P_{corr}(C) = \sum_{i=0}^n \alpha_i \cdot p^i (1-p)^{n-i}$$

**Example 6.8.**  $[n, k]$ -code from 6.5, coset leaders are 0000, 1000, 0100, 0010. Hence  $\alpha_0 = 1$ ,  $\alpha_1 = 3$ ,  $\alpha_2 = \alpha_3 = \alpha_4 = 0$ , so

$$P_{corr}(C) = (1-p)^4 + 3p(1-p)^3 = (1-p)^3(1+2p)$$

**Remark 6.9.** if  $d(C) = 2t+1$  or  $d(C) = 2t+2$ , then  $C$  can correct any  $t$  errors.

Hence every vector of weight  $\leq t$  is a coset leader and so  $\alpha_i = \binom{n}{i}$  for  $0 \leq i \leq t$ .

**Definition** (Rate).  $[n, k]$ -code  $C$  uses  $n$  symbols to send  $k$  message symbols; it has *rate*  $R(C) = k/n$ .

A good code will have a high rate.

**Definition** (Capacity). the *capacity*  $\mathcal{C}(p)$  of a binary symmetric channel with symbol error probability  $p$  is

$$\mathcal{C}(p) = 1 + p \cdot \log_2(p) + (1 - p) \cdot \log_2(1 - p)$$

**Theorem 6.12** (Shanon's theorem). channel binary symmetric with symbol error probability  $p$ .

Let  $R < \mathcal{C}(p)$ .

Then for any  $\epsilon > 0$ ,  $\exists$  for sufficient large  $n$ , an  $[n, k]$ -code  $C$  of rate  $k/n \geq R$  such that  $P_{err}(C) < \epsilon$ .

**Definition** ( $P_{symb}$ ).  $P_{symb}$  denotes the average probability that a message symbol is in error after decoding.

**Theorem 6.14.** let  $C$  a binary  $[n, k]$ -code and let  $A_i$  denote the number of codewords of  $C$  of weight  $i$ .

Then, if  $C$  is used for error detection, the probability of an incorrect message being undetected is

$$P_{undetec}(C) = \sum_{i=1}^n A_i \cdot p^i (1 - p)^{n-i}$$

**Example 6.15.** for the code of 6.5,

$$P_{undetec}(C) = \underbrace{1 \cdot p^2(1 - p)^2}_{0101} + \underbrace{2 \cdot p^3(1 - p)^{4-3=1}}_{1011, 1110} = p^2 - p^4$$

## 2.6 Exercises

**Exercise 5.1.** is the binary  $(11, 24, 5)$ -code linear?

*Proof.* recall: a  $q$ -ary linear  $[n, k, d]$ -code is also a  $q$ -ary  $(n, q^k, d)$ -code, but the opposite is not always true.

$\implies$   $(11, 24, 5)$ -code to be linear it would mean that  $24 = q^k$ .

Since it is binary,  $q = 2$ . But  $24 \neq 2^k$ , so it can not be linear.  $\square$

**Exercise 5.2.**  $E_n$  is the code of all even-weight vectors of  $V(n, 2)$ , which is linear. What are the parameters  $[n, k, d]$  of  $E_n$ ? Write a GM for  $E_n$  in standard form.

*Proof.*

$$E_n = \{x \in V(n, 2) | w(x) \text{ is even}\}$$

$\longrightarrow$  is the binary even-weight ode (the single parity-check code); it's linear since

- the zero vector has even weight

- over  $GF(2)$ , the sum of two even-weight vectors has even weight

Recall:  $F_q^n \iff V(n, q)$ , so  $V(n, 2) \cong F_2^n$

$$\implies E_n = \{(x_1, \dots, x_n) \in F_2^n \mid x_1 + \dots + x_n = 0\}$$

minimum distance is 2, since

- no word of weight 1 is in  $E_n$
- words of weight 2 are in  $E_n$

Therefore, parameters are  $[n, n-1, 2]$

GM:

$$G = [I_{n-1} \mid 1]$$

which generates vectors of the form

$$(a_1, \dots, a_{n-1}, a_1 + \dots + a_{n-1})$$

□

**Exercise 5.5.** Prove that in a *binary linear code*, either all the codewords have even weight or exactly half have even weight and half odd weight.

*Proof.* let  $C_e, C_o$  be subsets of  $C$  with even and odd weights respectively.

So that  $C = C_e \cup C_o$  (disjoint union).

For binary vectors  $x, y$ :

$$w(x + y) = w(x) + w(y) \pmod{2}$$

ie. over  $F_2$ , the parity of the weight is additive:

$$even + even = even$$

$$even + odd = odd$$

$$odd + odd = even$$

Now we have two possible cases:

- $C$  has no odd weighted codewords:  
 $C_o = \emptyset, \forall x \in C$   $x$  is even weighted
- $C$  has at least one odd-weighted codeword:  
let  $u \in C_o$ , consider the map

$$\begin{aligned} \phi : C_e &\longrightarrow C_o \\ c &\longmapsto c + u \end{aligned}$$

with  $c \in C_e, u \in C_o, \phi(c) \in C_o$ .

$\phi$  is bijective, since  $\phi(\phi(c)) = (c + u) + u = c$ ,

so  $\phi$  is its own inverse; hence  $|C_e| = |C_o|$ .

Since  $C = C_e \cup C_o$ , half codewords are even weighted and half odd.



□

**Exercise 6.1.** construct standard arrays for codes having each of the GMs:

$$G_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad G_2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

*Proof.* recall: for a binary code with GM  $G$ , the code  $C$  consists of all binary linear combinations of the rows of  $G$ .

For  $G_1$ :

rows:  $r_1 = (1, 0)$ ,  $r_2 = (0, 1)$ . A codeword is any linear combination:

$$uG_1 = a \cdot r_1 + b \cdot r_2, \quad \text{with } a, b \in \{0, 1\}$$

4 combinations:

$$0 \cdot r_1 + 0 \cdot r_2 = (0, 0)$$

$$1 \cdot r_1 + 0 \cdot r_2 = (1, 0)$$

$$0 \cdot r_1 + 1 \cdot r_2 = (0, 1)$$

$$1 \cdot r_1 + 1 \cdot r_2 = (1, 0) + (0, 1) = (1, 1)$$

$$\implies C_1 = \{00, 10, 01, 11\}$$

standard array: 00 01 10 11

For  $G_2$ :

rows:  $r_1 = (1, 0, 1)$ ,  $r_2 = (0, 1, 1)$ . Codeword:  $a \cdot r_1 + b \cdot r_2$  with  $a, b \in \{0, 1\}$ .

4 combinations:

$$0 \cdot r_1 + 0 \cdot r_2 = (0, 0, 0)$$

$$1 \cdot r_1 + 0 \cdot r_2 = (1, 0, 1)$$

$$0 \cdot r_1 + 1 \cdot r_2 = (0, 1, 1)$$

$$1 \cdot r_1 + 1 \cdot r_2 = (1, 0, 1) + (0, 1, 1) = (1, 1, 0)$$

$$\implies C_2 = \{000, 101, 011, 110\}$$

$\implies$  so the 1st row of the standard array is: 000 101 011 110.

Now, pick a non used vector: 001, add it to the code  $C_2$ :

$$001 + C_{2_i}$$

$$001 + 000 = 001$$

$$001 + 101 = 100$$

$$001 + 011 = 010$$

$$001 + 110 = 111$$

$\Rightarrow$  2nd row of the standard array: 001 100 010 111  
Thus the standard array is

$$\begin{array}{cccc} 000 & 101 & 011 & 110 \\ 001 & 100 & 010 & 111 \end{array}$$

ie.  $2^3 = 8$  vectors, a  $V(3, 2)$  vector space.  $\square$

**Exercise 6.4.**  $C$  a binary  $[n, k]$ -code with minimum distance  $2t + 1$  (or  $2t + 2$ ). Given that  $p$  is very small, show that an approximate value of  $P_{err}(C)$  is  $\left(\binom{n}{t+1} - \alpha_{t+1}\right)p^{t+1}$ , where  $\alpha_{t+1}$  is the number of coset leaders of  $C$  of weight  $t + 1$ .

*Proof.* From 6.9, if  $d(C) = 2t + 1$  (or  $2t + 2$ )  $\Rightarrow C$  can correct any  $t$  errors.

Hence every vector of weight  $\leq t$  is a coset leader and so  $\alpha_i = \binom{n}{i}$  for  $0 \leq i \leq t$ .

$$P_{corr}(C) = \sum_{i=0}^n \alpha_i \cdot p^i (1-p)^{n-i} \quad (1)$$

where  $\alpha_i$  is the number of coset leaders of weight  $i$

From the  $\binom{n}{i}$  vectors of weight  $i$ , exactly  $\alpha_i$  are coset leaders, so the non coset leaders are  $\binom{n}{i} - \alpha_i$ . This is when a decoding error happens.

$$\Rightarrow P_{err}(C) = \sum_{i=0}^n \left( \binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i}$$

Alternatively from 1:

$$P_{err}(C) = 1 - P_{corr}(C) = 1 - \sum_{i=0}^n \alpha_i \cdot p^i (1-p)^{n-i}$$

Notice that

$$\sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} = (p + (1-p))^n = 1^n = 1$$

thus

$$\begin{aligned} P_{err}(C) &= \sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} - \sum_{i=0}^n \alpha_i \cdot p^i (1-p)^{n-i} \\ &= \sum_{i=0}^n \left( \binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i} \end{aligned}$$

Now, from 6.9,  $\forall i \leq t$ ,  $\alpha_i = \binom{n}{i}$ , thus  $\binom{n}{i} - \alpha_i = 0$  for the first  $t$  terms in the sum.

Thus

$$\begin{aligned} P_{err}(C) &= \sum_{i=0}^n \left( \binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i} \\ &= \left( \binom{n}{t+1} - \alpha_{t+1} \right) p^{t+1} (1-p)^{n-t+1} + \dots \end{aligned}$$

where the  $\dots$  represent the terms with  $p^{t+2}$  and higher.

Then, since  $p$  is very small (by the exercise definition),  $(1-p)^{n-t+1} \equiv 1$ , and the other higher powers of  $p$  are negligible.

Hence,

$$P_{err}(C) = \left( \binom{n}{t+1} - \alpha_{t+1} \right) \cdot p^{t+1}$$

□

### 3 Dual code, PCM and Syndrome

#### 3.1 Dual code

**Definition** (Orthogonal). Inner product:

$$u \cdot v = u_1 v_1 + \dots + u_n \cdot v_n = \sum_{i=1}^n u_i v_i$$

If  $u \cdot v = 0$ , then  $u$  and  $v$  are called *orthogonal*.

**Lemma 7.1.** for any  $u, v, w \in V(n, q)$  and  $\lambda, \mu \in GF(q)$ ,

- i.  $u \cdot v = v \cdot u$
- ii.  $(\lambda u + \mu v) \cdot w = \lambda(u \cdot w) + \mu(v \cdot w)$

*Proof.* (at exercise 7.1). □

**Definition** (Dual code). given a linear  $[n, k]$ -code  $C$ , the *dual code* of  $C$ , denoted  $C^\perp$ , is defined to be the set of those vectors of  $V(n, q)$  which are orthogonal to every codeword of  $C$ . ie.

$$C^\perp = \{v \in V(n, q) \mid v \cdot u = 0 \forall u \in C\}$$

**Lemma 7.2.**  $C$  a  $[n, k]$ -code with GM  $G$ . Then  $v \in V(n, q)$  belongs to  $C^\perp$  iff  $v$  is orthogonal to every row of  $G$ , ie.

$$v \in C^\perp \iff v \cdot G^T = 0$$

*Proof.* let rows of  $G$  be  $r_1, \dots, r_k$ , suppose  $v \cdot r_i = 0 \ \forall i$  (ie.  $v$  orthogonal to every row of  $G$ ).

If  $u \in C$  (ie. a codeword of  $C$ ), then  $u = \sum_{i=1}^k \lambda_i r_i$  for some scalars  $\lambda_i$ , so that

$$v \cdot u = v \cdot \sum_{i=1}^k \lambda_i r_i = \sum_{i=1}^k \lambda_i \cdot \underbrace{(v \cdot r_i)}_0 = \sum \lambda_i \cdot 0 = 0$$

$\implies$  hence  $v$  is orthogonal to every codeword of  $C$ , and so is in  $C^\perp$ .  $\square$

**Theorem 7.3.**  $C$  a  $[n, k]$ -code over  $GF(q)$ . Then the dual code  $C^\perp$  of  $C$  is a linear  $[n, n - k]$ -code.

*Proof.* i. first show that  $C^\perp$  is a linear code:

let  $v_1, v_2 \in C^\perp$  and  $\lambda, \mu \in GF(q)$ . Then  $\forall u \in C$ ,

$$\begin{aligned} (\lambda v_1 + \mu v_2) \cdot u &= \lambda(v_1 \cdot u) + \mu(v_2 \cdot u) \\ &\text{notice that } v_1, v_2 \in C^\perp, \text{ so} \\ &\text{that } v_1, v_2 \text{ are orthogonal with } u \\ &\text{hence } (v_1 \cdot u) \text{ and } (v_2 \cdot u) \text{ are } 0 \\ &= \lambda \cdot 0 + \mu \cdot 0 = 0 \end{aligned}$$

Hence  $\lambda \cdot v_1 + \mu \cdot v_2 \in C^\perp$ , and so  $C^\perp$  is linear, since it is closed under linear combinations.

ii. Next, show that  $C^\perp$  has dimension  $n - k$ :

Let  $G = [g_{ij}]$  be a GM of  $C$ .

Then by Lemma 7.2, the elements of  $C^\perp$  are the vectors  $v = (v_1, v_2, \dots, v_n)$ , since  $v \in C^\perp \iff v \cdot G^T = 0$ , satisfying

$$\sum_{j=1}^m g_{ij} v_j = 0, \ \forall i = [k]$$

$\longrightarrow$  this is a system of  $k$  independent homogeneous equations in  $n$  unknowns, with solution space  $C^\perp$ . From linear algebra we know that it has dimension  $n - k$ . To see it:

if codes  $C_1$  and  $C_2$  are equivalent, then  $C_1^\perp$ ,  $C_2^\perp$  are also equivalent. Hence is enough to show that  $\dim(C^\perp) = n - k$ , where  $C$  is a standard form GM:

$$C = \begin{pmatrix} 1 & \dots & 0 & a_{1,1} & \dots & a_{1,n-k} \\ \vdots & & \vdots & \vdots & & \vdots \\ 0 & \dots & 1 & a_{k,1} & \dots & a_{k,n-k} \end{pmatrix}$$

Then

$$C^\perp = \{(v_1, \dots, v_n) \in V(n, q) \mid v_i + \sum_{j=1}^{n-k} a_{i,j} v_{k+j} = 0, i = 1, 2, \dots, k\}$$

For each of the  $q^{n-k}$  choices of  $(v_{k+1}, \dots, v_n)$ ,  $\exists! (v_1, v_2, \dots, v_n) \in C^\perp$ .

Hence  $|C^\perp| = q^{n-k}$ , therefore  $\dim(C^\perp) = n - k$ . □

**Theorem 7.5.** for any  $[n, k]$ -code  $C$ ,  $(C^\perp)^\perp = C$ .

*Proof.* clearly  $C \subseteq (C^\perp)^\perp$ , since every vector in  $C$  is orthogonal to every vector in  $C^\perp$ .

But  $\dim((C^\perp)^\perp) = n - (n - k) = k = \dim(C)$ , therefore  $C = (C^\perp)^\perp$ . □

### 3.2 PCM (parity-check matrix)

**Definition (PCM).** A *parity-check matrix* (PCM)  $H$  for a  $[n, k]$ -code  $C$  is a generator matrix of  $C^\perp$ .

$\implies H$  is a  $(n - k) \times n$  matrix satisfying  $GH^T = 0$ .

From 7.2 and 7.5,  
if  $H$  a PCM of  $C$ , then

$$C = \{x \in V(n, q) \mid xH^T = 0\}$$

$\implies$  Any linear code is completely specified by a PCM.

**Intuition .** i. PCM:  $G$  specifies how to build a codeword, the PCM  $H$  specifies the rules a vector must follow to be considered a valid codeword.

ii. dual-code: orthogonal "shadow", see  $C$  as a plane through 3D origin;  $C^\perp$  is the (normal vector) line emerging from the plane.

Rows of the PCM are the basis for the dual-code  $C^\perp \implies$  ie. the rules that define  $C$  are themselves a code.

**Theorem 7.6.** if  $G = [I_k | A]$  is the standard form generator matrix of an  $[n, k]$ -code  $C$ , then a PCM for  $C$  is  $H = [-A^T \mid I_{n-k}]$ .

eg.

$$G = \begin{bmatrix} 1 & \dots & 0 & a_{1,1} & \dots & a_{1,n-k} \\ \vdots & & \vdots & \vdots & & \vdots \\ 0 & \dots & 1 & a_{k,1} & \dots & a_{k,n-k} \end{bmatrix}$$

$$H = \begin{bmatrix} -a_{1,1} & \dots & -a_{1,n-k} & 1 & \dots & 0 \\ \vdots & & \vdots & \vdots & & \vdots \\ -a_{k,1} & \dots & -a_{k,n-k} & 0 & \dots & 1 \end{bmatrix}$$

which, in binary, the minus signs are unnecessary.

**Definition** (Standard form). a PCM that is  $H = [B \mid I_{n-k}]$

### 3.3 Syndrome

**Definition** (Syndrome). let  $H$  a PCM of a  $[n, k]$ -code  $C$ .

Then  $\forall y \in V(n, q)$ , the  $1 \times (n - k)$  row vector

$$S(y) = yH^T$$

is called the *syndrome* of  $y$ .

**Remark .** i. for  $h_1, h_2, \dots, h_{n-k}$  being rows of  $H$ , then

$$S(y) = (y \cdot h_1, y \cdot h_2, \dots, y \cdot h_{n-k})$$

ii.  $S(y) = 0 \iff y \in C$

**Intuition .** Syndrome is the "symptom" of the error. It ignores the valid part of the message and isolates the noise.

$$S(y) = yH^T = (c + e)H^T = \underbrace{cH^T}_0 + eH^T = 0 + eH^T = eH^T$$

Syndrome allows the receiver to identify what went wrong without knowing what was originally sent.

**Lemma 7.8.** two vectors  $u, v$  are the same coset of  $C$  iff they have the same syndrome.

*Proof.* if  $u, v$  in the same coset of  $C$

$$\begin{aligned} \implies u + C &= v + C \\ \implies u - v &\in C \\ \implies (u - v)H^T &= 0 \\ \implies uH^T &= vH^T \\ \implies S(u) &= S(v) \end{aligned}$$

□

**Corollary 7.9.** There is a one-to-one correspondence between cosets and syndromes.

—→ for small  $n$ , it's fast to locate the received vector  $y$  in the array, but for large  $n$ : use syndrome to find which coset (ie. which row of the array) contains  $y$ .

**Example 7.9.** (from example 6.5)

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

by 7.6,

$$H = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

Hence the syndromes of the coset leaders are

$$S(0000) = 00$$

$$S(1000) = 11$$

$$S(0100) = 01$$

$$S(0010) = 10$$

The standard array becomes:

| <i>coset leaders</i> |      |      |      | <i>syndromes</i> |
|----------------------|------|------|------|------------------|
| 0000                 | 1011 | 0101 | 1110 | 00               |
| 1000                 | 0011 | 1101 | 0110 | 11               |
| 0100                 | 1111 | 0001 | 1010 | 01               |
| 0010                 | 1001 | 0111 | 1100 | 10               |

so, decoding algorithm: when a vector  $y$  is received, calculate  $S(y) = yH^T$ , locate  $S(y)$  in the *syndromes* column of the array.

Locate  $y$  in the corresponding row and decode as the codeword at the top of the column containing  $y$ .

→ in practice: *syndrome lookup table*:

| syndrome $z$ | coset leader $f(z)$ |
|--------------|---------------------|
| 00           | 0000                |
| 11           | 1000                |
| 01           | 0100                |
| 10           | 0010                |

decoding procedure:

1. receive  $y$ , calculate  $S(y) = yH^T$
2. let  $z = S(y)$ , locate  $z$  in the first column of the lookup table
3. decode  $y$  as  $y - f(z)$

### 3.4 Incomplete decoding

For  $d(C) = 2t + 1$  (or  $2t + 2$ ), guarantee the correction of  $\leq t$  errors in any codeword and also detect some cases of more than  $t$  errors.

→ Arrange the cosets in order of increasing weight of the coset leaders.

Divide the array into a top part and a bottom part; top: cosets whose leaders have weights  $\leq t$ .

- If  $y$  in top part: decode as usual (assuming  $\leq t$  errors).
- If  $y$  in bottom part: conclude that  $> t$  errors have occurred, ask for retransmission.

### 3.5 Exercises

**Exercise 7.1.** prove Lemma 7.1

*Proof.* For any  $u, v, w \in V(n, q)$  and  $\lambda, \mu \in GF(q)$ ,

- i.  $u \cdot v = v \cdot u$

$$u \cdot v = \sum_{i=1}^n u_i \cdot v_i = \sum_{i=1}^n v_i \cdot u_i = v \cdot u$$

- ii.  $(\lambda u + \mu v) \cdot w = \lambda(u \cdot w) + \mu(v \cdot w)$

$$\begin{aligned} (\lambda u + \mu v) \cdot w &= \sum (\lambda u_i + \mu v_i) w_i \\ &= \sum (\lambda u_i w_i + \mu v_i w_i) \\ &= \sum (\lambda u_i w_i) + \sum (\mu v_i w_i) \\ &= \lambda \sum (u_i w_i) + \mu \sum (v_i w_i) \\ &= \lambda \cdot (u \cdot w) + \mu \cdot (v \cdot w) \end{aligned}$$

□

**Exercise 7.2.** prove that if  $E_n$  is the binary even weight code of length  $n$ , then  $E_n^\perp$  is the repetition code of length  $n$ .

*Proof.* from exercise 5.2:

$$E_n = \{x \in V(n, 2) \mid w(x) \text{ is even}\} = \{(x_1, \dots, x_n) \in F_2^n \mid x_1 + \dots + x_n = 0\}$$

GM of  $E_n$ :

$$G = \left[ I_{n-1} \mid \begin{array}{c} 1 \\ \vdots \\ 1 \end{array} \right]$$



thus, by 7.6, PCM is  $H = [1 \ 1 \ \cdots \ 1]$ .

If  $H$  is a PCM for  $C$ , then the rows of  $H$  generate  $C^\perp$ .

Therefore

$$E_n^\perp = \langle \underbrace{[1 \ 1 \ \cdots \ 1]}_{\text{generator}} \rangle$$

then it's codewords are:

$$0 \cdot (1, \dots, 1) = (0, \dots, 0)$$

$$1 \cdot (1, \dots, 1) = (1, \dots, 1)$$

$$\implies E_n^\perp = \{00 \dots 00, 11 \dots 1\}$$

ie. the repetition code of length  $n$ . □

**Exercise 7.4.** for a binary linear code with PCM  $H$ , show that the transpose of the syndrome of a received vector is equal to the sums of those columns of  $H$  corresponding to where the errors occurred.

*Proof.* Let  $e$  the error,  $y$  the vector received, and  $x$  the original codeword. Then  $y = x + e$ .

Then

$$S(y) = yH^T = (x + e)H^T = \underbrace{xH^T}_{\substack{0, \text{ since } H \\ \text{is PCM of } C \\ \text{and } x \in C}} + eH^T = eH^T$$

So,

$$S(y)^T = e^T H = \sum_{j=1}^n e_j \cdot H_j$$

where  $H_j$  is the  $j$ -th column of  $H$ . □

**Exercise 7.6.** Let  $C$  be the ternary linear code with GM

$$G = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 1 \end{bmatrix}$$

(a) Find a generator matrix in standard form.

(b) Find a parity check matrix (PCM) for  $C$  in standard form.

(c) Use syndrome decoding to decode the received vectors: 2122, 1201, 2222

*Proof.* (a)

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

thus  $G = \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$  is PCM in standard form.

(b) since  $G = \begin{bmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$  then

$$H = \begin{bmatrix} -2 & -2 & 1 & 0 \\ -2 & -1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

(c)

$$S(y) = yH^T = S(y) = (y \cdot h_1, y \cdot h_2).$$

w/  $h_i$  the rows of  $H$ .

compute the lookup table (syndrome table):

| $e$  | $S(e)$ |
|------|--------|
| 0000 | 00     |
| 1000 | 11     |
| 2000 | 22     |
| 0100 | 12     |
| 0200 | 21     |
| 0010 | 10     |
| 0020 | 20     |
| 0001 | 01     |
| 0002 | 02     |

left: weight-1 errors, since we're over  $F_3$ ; right: coset leaders for the 9 cosets.

i.  $y = 2121$ :

$$S(y) = \begin{bmatrix} 2 & 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = [5, 5] \equiv [2, 2].$$

$S(y) = [2, 2]$ , corresponds (in the table) to  $e = 2000$ .

$$\implies c = y - e = 2121 - 2000 = 0121$$

ii.  $y = 1201$

$$S(y) = \begin{bmatrix} 1 & 2 & 0 & 1 \end{bmatrix} H^T = [3, 6] = [0, 0].$$

corresponds to  $e = 0000$

$$\implies c = y - e = 1201$$

iii.  $y = 2222$

$$S(y) = \begin{bmatrix} 2 & 2 & 2 & 2 \end{bmatrix} H^T = [6, 8] = [0, 2].$$

corresponds to  $e = 0002$

$$\implies c = y - e = 2222 - 0002 = 2220$$

□

## 4 Hamming Codes

**Definition** (Binary Hamming Code). let  $r$  a positive integer,  $H$  a  $r \times (2^r - 1)$  matrix whose columns are the distinct non-zero vectors of  $V(r, 2)$ .

The code having  $H$  as its PCM is called a *binary Hamming code*, denoted  $Ham(r, 2)$ .

**Remark .**  $Ham(r, 2)$  has length  $n = 2^r - 1$  and dimension  $k = n - r$ .

Redundancy of the code:  $r = n - k$ , the number of check symbols in each codeword.

**Exercise 8.1.** i.  $r = 2$ :  $H = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

by Theorem 7.6,  $G = [1 \ 1 \ \underbrace{1}_I]$

so  $Ham(2, 2)$  is just the binary triple repetition code.

ii.  $r = 3$ : a PCM for  $Ham(3, 2)$  is

$$H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

(have taken columns in the natural order of increasing binary numbers from 1 to 7)

$H$  in standard form: rearrange columns:

$$H = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

by Theorem 7.6, a generator matrix for  $Ham(3, 2)$  is

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

**Theorem 8.2.** the binary Hamming code  $Ham(r, 2)$ , for  $r \geq 2$ ,

1. is a  $[2^r - 1, 2^r - 1 - r]$ -code

2. has minimum distance 3 (hence is single-error-correcting)
3. is a *perfect code*

**Lemma** (Decoding with a binary Hamming code).  $Ham(r, 2)$  is a perfect single-error-correcting code  $\implies$  the coset leader are precisely the  $2^r$  ( $= n + 1$ ) vectors of  $V(n, 2)$  of weight  $\leq 1$ .

Let  $j$ -th column of  $H$  be the binary representation of  $j$ . Decoding algorithm:

1. receive vector  $y$ , calculate its syndrome:  $S(y) = yH^T$
2. if  $S(y) = 0$ , assume  $y = c$  (original sent codeword)
3. if  $S(y) \neq 0$ , assuming a single error,  $S(y)$  gives the binary representation of the error position, and the error can be corrected.

**Example .**  $H = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$ , if  $y = 1101011$  then

$$S(y) = y \cdot H^T = 110$$

$\implies$  6-th position, and so we decode  $y$  as 1101001

## 4.1 Extended binary Hamming codes

$\hat{Ham}(r, 2)$ , obtained from  $Ham(r, 2)$  by adding an overall parity-check.

Not ideal for complete decoding, but good for incomplete decoding (correct any single error and detect any double error).

Let  $H$  be a PCM for  $Ham(r, 2)$ . A PCM  $\hat{H}$  for  $\hat{Ham}(r, 2)$ :

$$\hat{H} = \begin{bmatrix} & & & 0 \\ & H & & \vdots \\ & & & 0 \\ 1 & \cdots & 1 & 1 \end{bmatrix}$$

If  $H$  is taken with columns in increasing order of binary numbers, we have a neat incomplete decoding algorithm:

**Example 8.3.**  $r = 3$ ,  $\hat{Ham}(3, 2)$ . Received  $y$ ,  $S(y) = y\hat{H}^T = (s_1, s_2, s_3, s_4)$

- i. if  $s_4 = 0$  and  $(s_1, s_2, s_3) = 0$ : assume no errors
- ii. if  $s_4 = 0$  and  $(s_1, s_2, s_3) \neq 0$ : assume  $\geq 2$  errors and ask retransmission
- iii. if  $s_4 = 1$  and  $(s_1, s_2, s_3) = 0$ : assume 1 error in the last position

- iv. if  $s_4 = 1$  and  $(s_1, s_2, s_3) \neq 0$ : assume 1 error in the  $j$ -th position, where  $j$  is the number binary-represented by  $(s_1, s_2, s_3)$

**Theorem 8.4.** suppose  $C$  a linear  $[n, k]$ -code over  $GF(q)$  with PCM  $H$ . Minimum distance of  $C$  is  $d$  iff any  $d - 1$  columns of  $H$  are linearly independent but some  $d$  columns are linearly dependent.

*Proof.* by 5.2, min. distance of  $C$  (ie.  $d(C)$ ) = smallest of the weights of the non-zero codewords.

Let  $x = x_1x_2 \dots x_n \in V(n, q)$ , then

$$\begin{aligned} x \in C &\iff xH^T = 0 \\ &\iff x_1H_1 + x_2H_2 + \dots + x_nH_n = 0 \end{aligned}$$

where  $H_i$  denote the columns of  $H$ .

$\implies$  each codeword  $x$  of weight  $d$  corresponds with a set of  $d$  linearly dependent columns of  $H$ .

On the other hand, if  $\exists$  a set of  $d - 1$  lin.dependent columns of  $H$ , say  $H_{i_1}, H_{i_2}, \dots, H_{i_{d-1}}$ , then  $\exists x_{i_1}, \dots, x_{i_{d-1}}$  scalars, not all zero, such that

$$x_{i_1}H_{i_1} + x_{i_2}H_{i_2} + \dots + x_{i_{d-1}}H_{i_{d-1}} = 0$$

But then, the vector

$$x = (0 \dots 0x_{i_1}0 \dots x_{i_2}0 \dots x_{i_{d-1}}0 \dots 0)$$

having  $x_{i_j}$  in the  $j$ -th position for  $j \in [d - 1]$ , would satisfy  $xH^T = 0$ , and so would be a non-zero codeword of weight  $< d$ .  $\square$

## 4.2 q-ary Hamming codes

For  $C$  a linear code with  $d(C) = 3$  (min.dist), any 2 columns of a PCM  $H$  must be linearly independent

$\implies$  columns of  $H$  must be non-zero, and no column can be a scalar multiple of another

Any  $0 \neq v \in V(r, q)$  vector has  $q - 1$  non-zero scalar multiples, forming the set

$$\{\lambda v \mid \lambda \in GF(q), \lambda \neq 0\}$$

$\longrightarrow$  the  $q^r - 1$  non-zero vectors of  $V(r, q)$  may be partitioned into  $\frac{q^r - 1}{q - 1}$  such sets, called *classes*.

$\longrightarrow$  two vectors are scalar multiples of each other if they are in the same class

By choosing one vector from each class, obtain a set of  $\frac{q^r-1}{q-1}$  vectors, where not two are linearly dependent.

By Theorem 8.4, taking these as columns of  $h$  gives a PCM for a

$$\left[ \frac{q^r-1}{q-1}, \frac{q^r-1}{(q-1)-r}, 3 \right] \text{-code}$$

$\implies$  called  $q$ -ary Hamming code,  $Ham(r, q)$ .

$\longrightarrow$  PCM is unique up to permutation of columns

$\implies$  Hamming codes are linear codes

**Theorem 8.6.**  $Ham(r, q)$  is a perfect single-error-correcting code.

*Proof.*  $Ham(r, q)$  is constructed to be a  $(n, M, 3)$ -code, with  $n = \frac{q^r-1}{q-1}$  and  $M = q^{n-r}$ .

With  $t = 1$ , the LHS of the sphere-packing bound becomes

$$q^{n-r} \cdot (1 + n(q-1)) = q^{n-r} \cdot (1 + q^r - 1) = q^n$$

which is the RHS, so  $Ham(r, q)$  is a perfect code.  $\square$

**Decoding  $q$ -ary Hamming code** Since Ham is perfect single-error correcting code, the coset leaders, other than 0, are the vectors of weight 1.

Syndrome of such coset leader:

$$S(0 \dots 0 b 0 \dots 0) = (0 \dots 0 b 0 \dots 0) \cdot H^T = b \cdot H_j^T$$

where  $H_j$  is the  $j$ -th column of  $H$ , and  $b$  is at the  $j$ -th position of  $y$ .

Decode:

- given received  $y$ , calculate  $S(y) = yH^T$
- if  $S(y) = 0$ , assume no errors
- if  $S(y) \neq 0$ , then  $S(y) = b \cdot H_j^T$  for some  $b$  and  $j$   
correct by: subtract  $b$  from the  $j$ -th entry of  $y$ .

Example:

$$q = 5, H = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 3 & 4 \end{bmatrix}, y = 203031$$

Then,

$$\begin{aligned} S(y) &= y \cdot H^T = 2 \cdot H_1^T + 3 \cdot H_3^T + 3 \cdot H_5^T + 1 \cdot H_6^T \\ &= 2 \cdot [0 \ 1] + 3 \cdot [1 \ 1] + 3 \cdot [1 \ 3] + 1 \cdot [1 \ 4] \\ &= [0 \ 2] + [3 \ 3] + [3 \ 9] + [1 \ 4] \\ &= [7 \ 8] = [2 \ 3] = \underbrace{2}_b \cdot \underbrace{[1 \ 4]}_{H_6^T} \pmod{5} \end{aligned}$$

$\implies$  error is in position 6-th, with magnitude  $b = 2$   
 $\implies$  subtract  $b$  from the  $j$ -th entry of  $y$ :  $1 - 2 \pmod{5} = -1 \pmod{5} = 4$ ,  
therefore  $y' = 203034$ .

## References

- [1] Raymond Hill. A First Course in Coding Theory, 1986.
- [2] J.H. van Lint. Introduction to Coding Theory, 1981.